

Juan Ramon Ortiz

juanortiz@proton.me | github.com/juanortiz | linkedin.com/in/juanortiz

SUMMARY

Electrical engineer with 5+ years of industrial automation experience transitioning into embedded systems and IoT software engineering. Hands-on experience in C/C++ systems programming, hardware integration, and serial communications. Pursuing an M.S. in Electrical Engineering specializing in Embedded Systems at Cal Poly Pomona.

SKILLS SUMMARY

Languages:	C, C++, Python, PowerShell, Rust, LaTeX
Embedded & Systems:	Serial Protocols (UART/SPI/I2C), Win32 API, Memory Optimization, GCC/g++, Cargo
Dev Environment:	Linux CLI, Git, QEMU/KVM, SSH, Helix, Zellij, VS Code
Hardware & Test:	Oscilloscopes, DMM, Electrical Soldering and Wire Troubleshooting
Industry Tools:	Logix Designer SDK, ControlFlash+ API, Studio 5000
Soft Skills:	Leadership, Project Management, Technical Documentation, Team Collaboration

PROJECTS

- EMG-Controlled Prosthetic Hand:** Developed embedded control software on a Teensy 4.1 to drive a robotic prosthetic hand using electromyography signals. Implemented a second-order Chebyshev filter in C for noise reduction, and designed a matrix-based finger position controller with per-finger state tracking. Collaborated on hardware signal conditioning circuit for EMG capture. Tech: C, Teensy 4.1, Arduino IDE, Signal Processing, EMG Sensing
- RP2040 Cyberdeck Controller:** Built a portable USB HID controller on a Pro Micro RP2040 integrating an M5Stack CardKB over I2C, analog joystick mouse control, and programmable macro buttons. Developed firmware in C++ using PlatformIO and the Arduino framework. Designed custom PCB schematic in KiCad and ordered fabricated prototype boards. Tech: C++, RP2040, USB HID, I2C, KiCad, PlatformIO, VS Code
- Automated PLC Firmware Upgrade Tool:** Built a Python/C# tool interfacing with the ControlFlash+ API and Logix Designer SDK to automate firmware flashing and disaster recovery across multiple PLC controllers, reducing process time by 70%. Tech: Python, C#, Logix Designer SDK, ControlFlash+ API
- TCP Industrial Logging Tool:** Built a Python CLI tool that connects via TCP to Linux-based industrial control servers, querying and logging system and controls data by target IP. Tech: Python, TCP/IP, Sockets, PowerShell, Linux
- Minimal OS Memory Optimization:** Reduced a Windows 11 VM to 806MB idle RAM on QEMU/KVM by auditing and disabling kernel services, implementing process lifecycle management via native Win32 API and Rust utilities. Tech: C++, Rust, Win32 API, PowerShell, QEMU/KVM
- Image Segmentation via DCT Feature Extraction:** Implemented a Bayesian MAP classifier in MATLAB to segment foreground and background in images using 8x8 block DCT features, zig-zag scanning, and class-conditional probability estimation. Tech: MATLAB, Signal Processing, Statistical Classification

EXPERIENCE

United Parcel Service (UPS) <i>Corporate PLC Obsolescence Supervisor (Hybrid)</i>	Ontario, California, USA <i>April 2024 - Present</i>
Project Leadership and PLC Programming: Led the HSLA conversion project to address obsolete controllers running on serial communications. Developed and modified standard PLC programs to resolve compatibility issues. Created a Vijeo Designer HMI program for legacy HMIS5T Magelis Panels to ensure compatibility with Studio 5000. Legacy Equipment Upgrades: Upgraded 200+ 1768-L45 controllers to CompactLogix 5069-L320ER nationwide, addressing PLC obsolescence and eliminating unsupported equipment that posed security risks. Escalation and Technical Support: Provide fourth-level escalation support for discontinued PLC platforms, resolving ladder logic, I/O, and communications failures through remote and onsite troubleshooting.	
<i>Automation Engineering Specialist (Promoted Internally)</i>	<i>October 2021 - April 2024</i>
Network Troubleshooting: Led the remapping of the automated network to ensure reliable connectivity by redrawing the ring network in AutoCAD Electrical. Sensor Integration: Integrated jam detection sensors into conveyor systems by wiring directly to existing I/O modules, achieving real-time fault detection without modifying PLC logic.	

EDUCATION

California Polytechnic State University <i>Master in Electrical Engineering (M.S.) Specialization: Embedded Systems</i>	Pomona, California, USA <i>(In Progress)</i>
Courses: Advanced Robotics, Nanoelectronics (Upcoming)	
<i>Bachelor in Electrical Engineering (B.S.) Specialization: Control Systems and Robotics</i>	<i>(Completed)</i>
Courses: Microcontrollers, Signals and Systems, RF Electronics, Advanced Control Systems	

CERTIFICATIONS

Rockwell Automation Learning+: ControlLogix Fundamentals, Ladder Logic (L1-L4), Maintenance, STL/SFC, Function Block Programming